

Algebra 1
Unit 11
Lesson 4 Practice

Name **Answer Key** _____
Date _____
Period _____

Use the scenario below for questions 1-5.

Janice started a bank account for her daughter with \$625, which earns 4.75% interest per year. The amount in her daughters account after x years can be modeled by the function $V(x) = \$625(1.0475)^x$.

1. Does the equation $V(x) = \$625(1.0475)^x$ represent exponential growth or decay?

exponential growth since $b > 1$

2. What is the y -intercept and what does it represent in this situation?

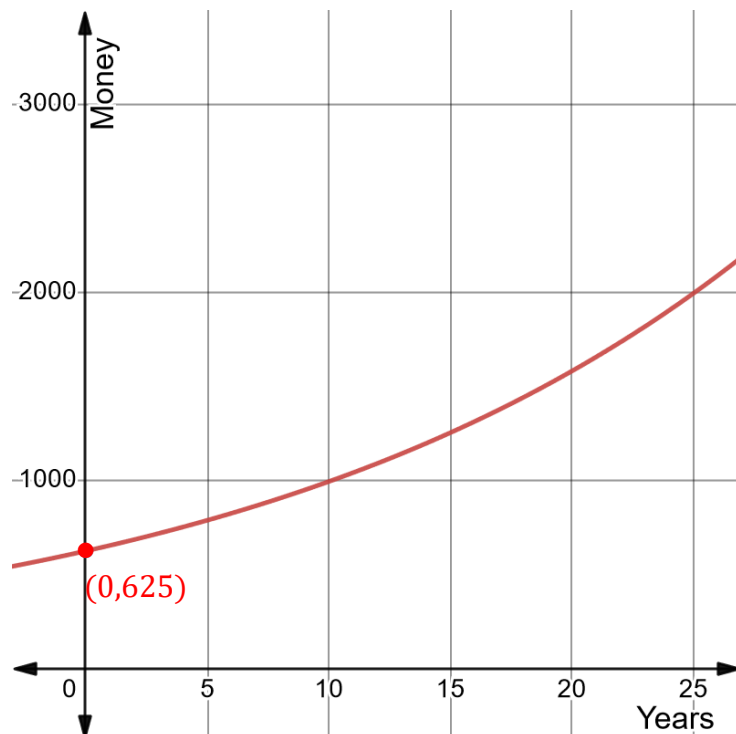
y -intercept is 625. This represents the initial amount deposited.

3. Explain why the value of b is not 4.75, the interest rate.

The interest rate is a percent. $4.75\% = \frac{4.75}{100} = .0475$.

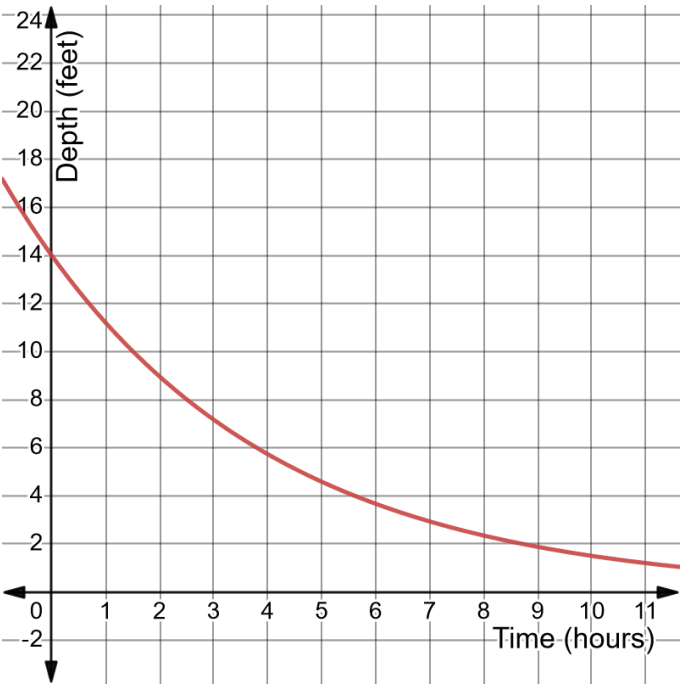
This is then added to the 100% which was deposited originally.

4. Graph the equation $V(x) = \$625(1.0475)^x$.



The equation $y = 14\left(\frac{4}{5}\right)^x$ represents the depth of the water as a pool drains in feet in x hours. Graph the function below, then determine the key features and interpret how each relates to the context.

5.



Key Features		Interpretations
6. Domain	$x \geq 0$	The domain is the time the pool drains for
7. Range	$0 \leq y \leq 14$	At time 0 hours, the pool is 14 feet deep until no water is left
8. y -intercept	$(0,14)$	The initial depth of the water in the pool
9. Constant Percent Rate of Change	$y = 14(1 - .2)^x$ $14(.8)^x$	Rate at which the pool is draining every hour
10. End Behavior	$x \rightarrow \infty, y \rightarrow 0$	As time continues, the pool drains until no water is left.